

Does Gender Disparity Exist Across Departments in Terms of Salary and Bonuses?

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Abstract

Gender-based disparities in the workplace have long been a critical issue in labor economics and workplace equality discussions. This study investigates how gender disparities in base pay and bonuses manifest across different departments. Using a Glassdoor dataset, we analyze base salary and bonus differences between men and women while controlling for factors such as age, education, seniority, and department. The findings help identify departments with the most significant gender pay gaps and provide actionable insights for reducing these disparities.

Research Question

How does gender disparity manifest across different departments in terms of base pay and bonuses?

Hypothesis

Men achieve higher base pay and bonuses compared to women across departments, even after controlling for age, education, and seniority.

Descriptive Statistics

About the Dataset

The dataset contains employee-level data, including salary, bonus, job title, gender, and departmental information. It allows for a detailed analysis of gender-based salary disparities.

Description of Attributes

Variable	Description	Unit/Category
JobTitle	Job title of the employee	Categorical
Gender	Gender of the employee	Male/Female
Age	Age of the employee	Years
PerfEval	Performance evaluation score	Integer (1 to 5)
Education	Highest educational attainment	College/Masters/PhD
Dept	Department where the employee works	Categorical (e.g., Sales, Admin)
Seniority	Seniority level of the employee	Integer
BasePay	Base salary of the employee	USD
Bonus	Bonus received by the employee	USD

Importing Data

```
library(tidyverse)

# Load the dataset
data <- read_csv("C:/Users/rayaa/Desktop/Fall 2024/Quatitative analysis/603
  ↳ final project/data/Glassdoor Gender Pay Gap.csv")

# Display first few rows
head(data)
```

```
# A tibble: 6 x 9
  JobTitle      Gender  Age PerfEval Education Dept Seniority BasePay Bonus
  <chr>         <chr> <dbl>   <dbl> <chr>   <chr>   <dbl>   <dbl> <dbl>
1 Graphic Designer Female   18     5 College Oper~     2   42363  9938
2 Software Engine~ Male    21     5 College Mana~     5  108476 11128
3 Warehouse Assoc~ Female   19     4 PhD      Admi~     5   90208  9268
4 Software Engine~ Male    20     5 Masters Sales     4  108080 10154
5 Graphic Designer Male    26     5 Masters Engi~     5   99464  9319
6 IT           Female   20     5 PhD      Oper~     4   70890 10126
```

Data Preprocessing

```
# Ensure proper data types
data$Gender <- as.factor(data$Gender)
data$Education <- as.factor(data$Education)
data$Dept <- as.factor(data$Dept)

# Check for missing values
sum(is.na(data))
```

```
[1] 0
```

Data Summarization

```
library(knitr)

# Summarize numeric variables
numeric_summary <- data %>% select_if(is.numeric) %>% summary()
kable(numeric_summary, caption = "Summary of Numeric Variables")
```

Table 2: Summary of Numeric Variables

Age	PerfEval	Seniority	BasePay	Bonus
Min. :18.00	Min. :1.000	Min. :1.000	Min. : 34208	Min. : 1703
1st Qu.:29.00	1st Qu.:2.000	1st Qu.:2.000	1st Qu.: 76850	1st Qu.: 4850
Median :41.00	Median :3.000	Median :3.000	Median : 93328	Median : 6507
Mean :41.39	Mean :3.037	Mean :2.971	Mean : 94473	Mean : 6467
3rd Qu.:54.25	3rd Qu.:4.000	3rd Qu.:4.000	3rd Qu.:111558	3rd Qu.: 8026
Max. :65.00	Max. :5.000	Max. :5.000	Max. :179726	Max. :11293

```
# Summarize categorical variables
categorical_summary <- data %>% select_if(is.factor) %>%
  summarise_all(~ paste(unique(.), collapse = ", "))
kable(categorical_summary, caption = "Summary of Categorical Variables")
```

Table 3: Summary of Categorical Variables

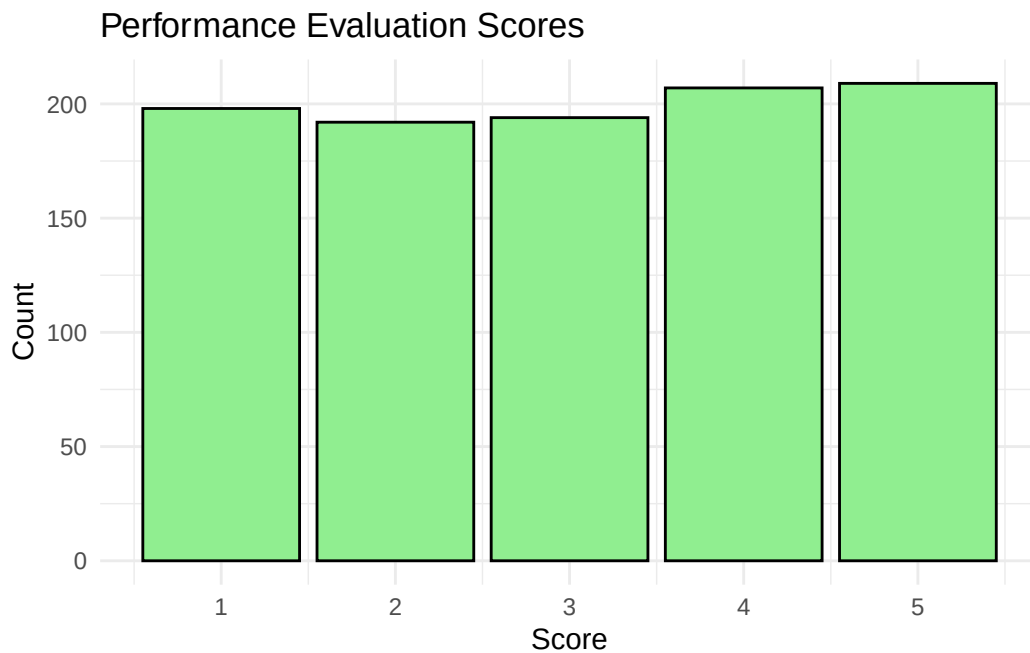
Gender	Education	Dept
Female, Male	College, PhD, Masters, High School	Operations, Management, Administration, Sales, Engineering

Visualizing the Data

Univariate Analysis

Performance Evaluation Scores

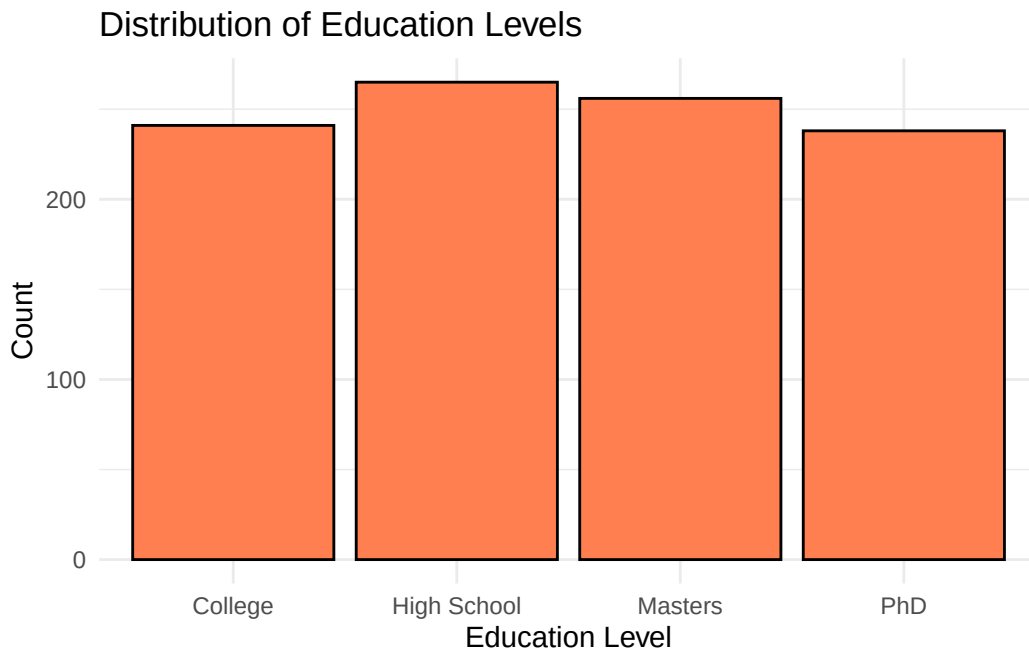
```
ggplot(data, aes(x = PerfEval)) +
  geom_bar(fill = "lightgreen", color = "black") +
  theme_minimal() +
  labs(title = "Performance Evaluation Scores", x = "Score", y = "Count")
```



The **Performance Evaluation Scores** graph shows a roughly uniform distribution of scores ranging from 1 to 5, with each score being represented by approximately the same number of employees (around 200). This suggests that performance evaluations are evenly distributed across the workforce, indicating no clear skew or bias toward any specific score.

Education Levels

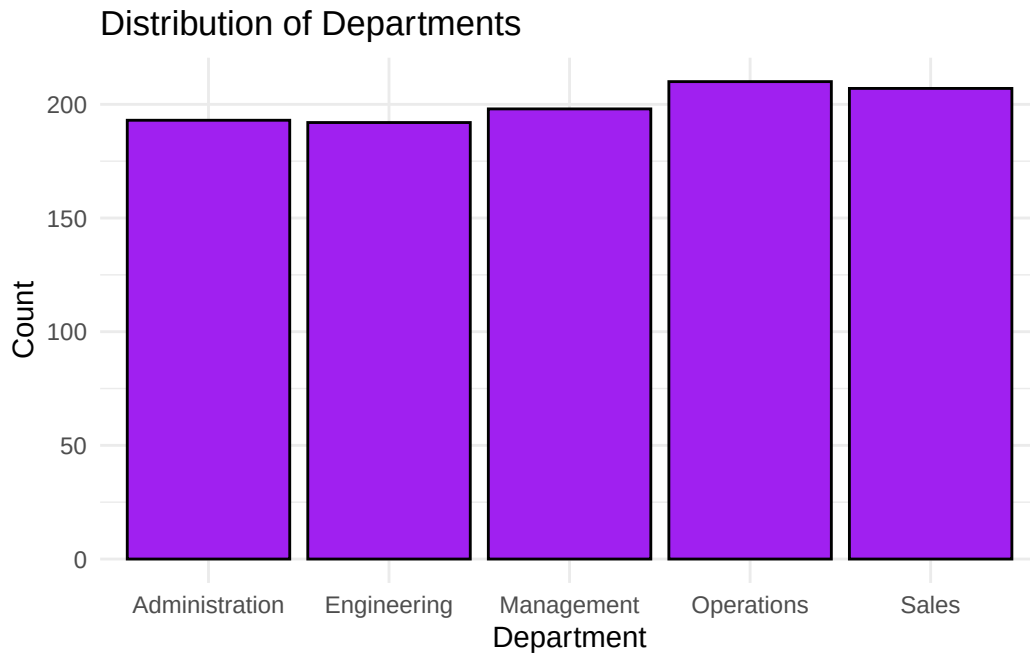
```
ggplot(data, aes(x = Education)) +  
  geom_bar(fill = "coral", color = "black") +  
  theme_minimal() +  
  labs(title = "Distribution of Education Levels", x = "Education Level", y =  
    ↪ "Count")
```



The **Distribution of Education Levels** graph illustrates the count of employees across four education categories: **High School, College, Masters, and PhD**. High School and Masters' degree holders form the largest groups, while College and PhD levels have slightly fewer employees. This balanced distribution indicates diversity in educational attainment within the workforce, with a notable presence of employees at various qualification levels.

Departments

```
ggplot(data, aes(x = Dept)) +  
  geom_bar(fill = "purple", color = "black") +  
  theme_minimal() +  
  labs(title = "Distribution of Departments", x = "Department", y = "Count")
```



The **Distribution of Departments** graph shows that employees are evenly spread across the five departments: **Administration, Engineering, Management, Operations, and Sales**, with each department having approximately 200 employees. This balanced distribution ensures that departmental comparisons are not biased by unequal sample sizes, providing a robust foundation for analyzing gender disparities in base pay and bonuses.

Graphs and Charts

Gender Disparity by Department

Base Pay Disparity

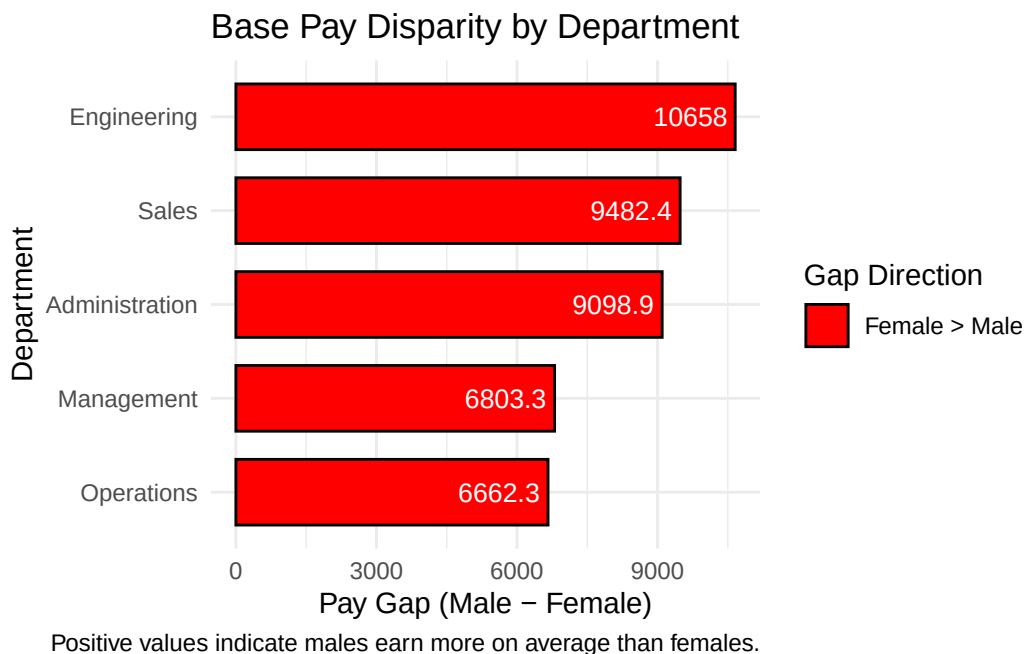
```
pay_gap <- data %>%
  group_by(Dept, Gender) %>%
  summarise(Avg_BasePay = mean(BasePay, na.rm = TRUE)) %>%
  pivot_wider(names_from = Gender, values_from = Avg_BasePay) %>%
  mutate(Gap = Male - Female)

ggplot(pay_gap, aes(x = reorder(Dept, Gap), y = Gap, fill = Gap > 0)) +
  geom_bar(stat = "identity", width = 0.7, color = "black") +
  coord_flip() +
```

```

theme_minimal() +
labs(
  title = "Base Pay Disparity by Department",
  x = "Department",
  y = "Pay Gap (Male - Female)",
  caption = "Positive values indicate males earn more on average than
    ↪ females."
) +
geom_text(aes(label = round(Gap, 1)), hjust = 1.1, color = "white", size =
  ↪ 3.5) +
scale_fill_manual(
  values = c("red", "blue"),
  name = "Gap Direction",
  labels = c("Female > Male", "Male > Female")
)

```

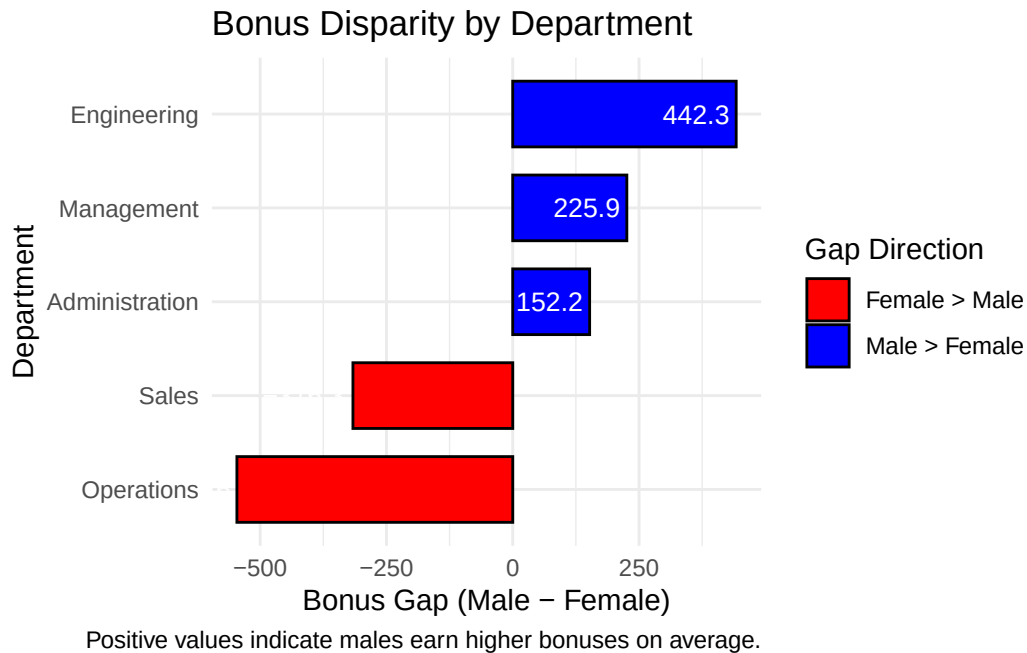


The **Base Pay Disparity by Department** graph highlights significant gender disparities in base pay across departments, where men earn more than women. The **Engineering** department shows the largest pay gap of approximately **10,658 USD**, followed by **Sales** (9,482 USD) and **Administration** (9,098 USD). Departments like **Management** and **Operations** exhibit smaller, yet notable, disparities. This graph underscores that the pay gap is persistent across all departments, with more pronounced differences in technical and sales-focused roles.

Bonus Disparity

```
bonus_gap <- data %>%
  group_by(Dept, Gender) %>%
  summarise(Avg_Bonus = mean(Bonus, na.rm = TRUE)) %>%
  pivot_wider(names_from = Gender, values_from = Avg_Bonus) %>%
  mutate(Gap = Male - Female)

ggplot(bonus_gap, aes(x = reorder(Dept, Gap), y = Gap, fill = Gap > 0)) +
  geom_bar(stat = "identity", width = 0.7, color = "black") +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Bonus Disparity by Department",
    x = "Department",
    y = "Bonus Gap (Male - Female)",
    caption = "Positive values indicate males earn higher bonuses on
      ↪ average."
  ) +
  geom_text(aes(label = round(Gap, 1)), hjust = 1.1, color = "white", size =
    ↪ 3.5) +
  scale_fill_manual(
    values = c("red", "blue"),
    name = "Gap Direction",
    labels = c("Female > Male", "Male > Female")
  )
)
```



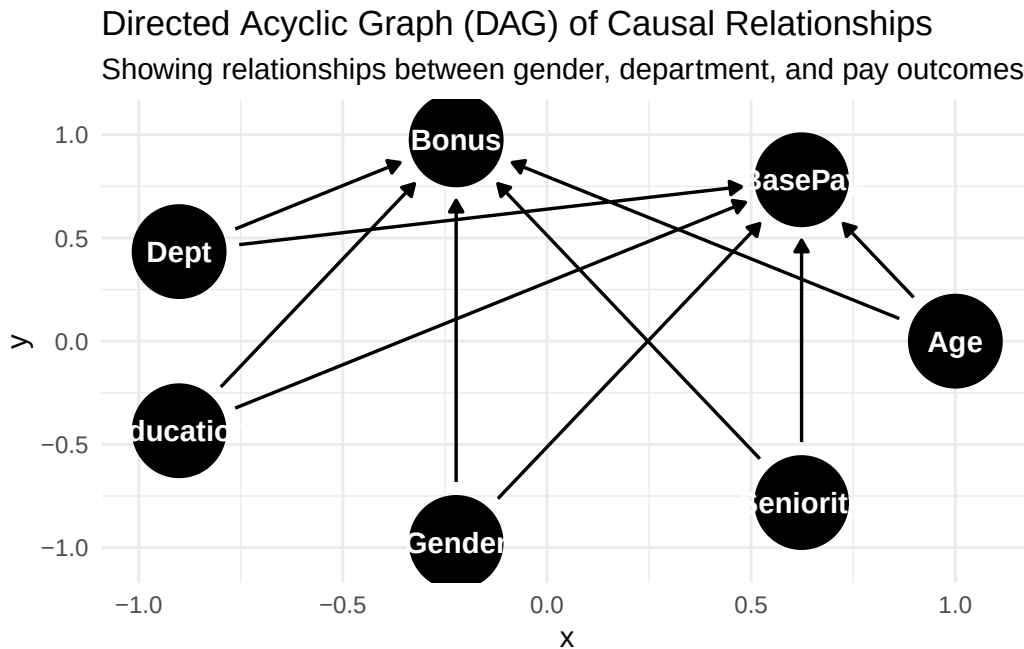
The **Bonus Disparity by Department** graph illustrates differences in average bonuses between males and females across departments. Positive values (in **blue**) indicate that males earn higher bonuses, while negative values (in **red**) indicate females earn higher bonuses.

- **Engineering**, **Management**, and **Administration** departments show higher bonuses for males, with the largest gap in **Engineering** (442.3 USD).
- Conversely, in **Sales** and **Operations**, females earn higher bonuses on average, with **Operations** showing the largest gap in favor of females (around -500 USD).

This mixed pattern suggests that bonus disparities are department-specific, with some departments favoring males and others favoring females.

Directed Acyclic Graph (DAG)

Causal Relationships Using DAG



Explanation of the DAG

The **Directed Acyclic Graph (DAG)** represents the relationships between various factors that influence **BasePay** and **Bonus**:

1. Outcome Variables:

- **BasePay** and **Bonus** are the primary focus, as they represent the outcomes of interest.
- Arrows point **toward** these variables, indicating the factors that are associated with them.

2. Primary Variable of Interest:

- **Gender** is the key variable being analyzed to understand its association with **BasePay** and **Bonus**.

3. Other Contributing Factors:

- **Dept (Department)**, **Age**, **Education**, and **Seniority** are additional variables that may help explain variations in **BasePay** and **Bonus**.
- Arrows from these factors point to **BasePay** and **Bonus**, suggesting their influence.

4. Relationships Depicted:

- Arrows from **Gender** to **BasePay** and **Bonus** indicate that gender is associated with differences in salary and bonus outcomes.
- Additional arrows from **Dept**, **Education**, **Seniority**, and **Age** show how these factors contribute to the observed variation.

5. Purpose of the DAG:

- The DAG helps to map out the relationships between variables in the analysis.
- It ensures that important variables, such as **Education** and **Seniority**, are considered when examining the association between **Gender** and the outcomes.

Summary:

The DAG visually organizes the variables and their relationships, showing how **BasePay** and **Bonus** can be associated with **Gender** while accounting for other factors like **Department**, **Age**, **Education**, and **Seniority**.

Model Fitting and Hypothesis Testing

Linear Regression Model

```
model <- lm(BasePay ~ Gender * Dept + Age + Education + Seniority, data =
  ↪ data)
summary(model)
```

Call:

```
lm(formula = BasePay ~ Gender * Dept + Age + Education + Seniority,
    data = data)
```

Residuals:

```
Min      1Q  Median      3Q      Max
```

-39501 -10242 -963 9234 48569

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	14240.66	2445.68	5.823	7.83e-09	***
GenderMale	8304.49	2152.37	3.858	0.000122	***
DeptEngineering	3010.14	2204.80	1.365	0.172481	
DeptManagement	5417.62	2215.29	2.446	0.014637	*
DeptOperations	862.88	2164.80	0.399	0.690278	
DeptSales	3548.73	2136.14	1.661	0.096975	.
Age	1017.96	33.14	30.714	< 2e-16	***
EducationHigh School	-1833.91	1331.57	-1.377	0.168749	
EducationMasters	4307.59	1347.47	3.197	0.001434	**
EducationPhD	6211.05	1368.83	4.537	6.39e-06	***
Seniority	9534.29	342.42	27.844	< 2e-16	***
GenderMale:DeptEngineering	2290.04	3053.85	0.750	0.453503	
GenderMale:DeptManagement	-195.52	3032.14	-0.064	0.948599	
GenderMale:DeptOperations	-738.45	2989.57	-0.247	0.804952	
GenderMale:DeptSales	5281.71	2998.21	1.762	0.078444	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14920 on 985 degrees of freedom

Multiple R-squared: 0.6582, Adjusted R-squared: 0.6533

F-statistic: 135.5 on 14 and 985 DF, p-value: < 2.2e-16

```
library(broom)

model_results <- tidy(model, conf.int = TRUE)

library(knitr)
kable(model_results, digits = 3, caption = "Linear Regression Results for
↪ Base Pay")
```

Table 4: Linear Regression Results for Base Pay

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	14240.664	2445.680	5.823	0.000	9441.321	19040.007
GenderMale	8304.491	2152.370	3.858	0.000	4080.733	12528.248

term	estimate	std.error	statistic	p.value	conf.low	conf.high
DeptEngineering	3010.136	2204.798	1.365	0.172	-	7336.777
					1316.505	
DeptManagement	5417.619	2215.292	2.446	0.015	1070.383	9764.854
DeptOperations	862.880	2164.802	0.399	0.690	-	5111.033
					3385.274	
DeptSales	3548.735	2136.145	1.661	0.097	-643.183	7740.652
Age	1017.958	33.144	30.714	0.000	952.918	1082.998
EducationHigh School	-	1331.574	-1.377	0.169	-	779.141
	1833.906				4446.953	
EducationMasters	4307.587	1347.467	3.197	0.001	1663.352	6951.822
EducationPhD	6211.050	1368.831	4.537	0.000	3524.890	8897.210
Seniority	9534.293	342.423	27.844	0.000	8862.331	10206.255
GenderMale:DeptEngineering	2290.035	3053.850	0.750	0.454	-	8282.835
					3702.765	
GenderMale:DeptManagement	1195.521	3032.141	-0.064	0.949	-	5754.677
					6145.719	
GenderMale:DeptOperations	738.455	2989.572	-0.247	0.805	-	5128.207
					6605.117	
GenderMale:DeptSales	5281.709	2998.213	1.762	0.078	-601.910	11165.327

Linear Regression Results for Base Pay

The table presents the results of a multiple linear regression model, where **BasePay** is the dependent variable. Below is a detailed explanation of the results:

Key Terms and Coefficients

1. Intercept:

- The baseline value for **BasePay** is **14,240.66 USD**.
- This represents the average salary for females in the **Administration department** when all other predictors (Age, Education, Seniority) are at their reference levels.

2. GenderMale:

- Males earn **8,304.49 USD** more than females on average, with a highly significant p-value (**< 0.001**).

- The confidence interval ([4,080.73, 12,528.25]) confirms a robust effect.

3. Department Effects:

- **DeptManagement:** Employees in Management earn **5,417.62 USD** more than those in Administration (**p = 0.015**), a statistically significant effect.
- **DeptSales:** Employees in Sales earn **3,548.74 USD** more, but this effect is marginally insignificant (**p = 0.097**).
- **DeptEngineering** and **DeptOperations** show no statistically significant differences from Administration.

4. Age:

- Each additional year of age increases **BasePay** by **1,017.96 USD** (**p < 0.001**), indicating a strong and consistent positive relationship.

5. Education:

- Compared to a **College** degree:
 - **Master's degree** increases pay by **4,307.59 USD** (**p = 0.001**).
 - **PhD** increases pay by **6,211.05 USD** (**p < 0.001**).
- **High School** has a negative but insignificant effect.

6. Seniority:

- **Seniority** has a significant positive effect (**9,534.29 USD**, **p < 0.001**), meaning each unit increase in seniority level substantially increases pay.

7. Interaction Terms (Gender * Dept):

- **GenderMale:DeptSales:** Males in Sales earn an additional **5,281.71 USD** relative to females in Sales, but this effect is only marginally significant (**p = 0.078**).
- Other interaction terms (e.g., GenderMale with Engineering, Management, Operations) are not statistically significant, suggesting no additional differences between genders in those departments.

Model Fit and Significance

- **Residual Standard Error: 14,920 USD**, indicating the average deviation of predicted values from actual base pay.
 - **Multiple R-squared: 0.6582** — the model explains **65.8%** of the variance in base pay.
 - **Adjusted R-squared: 0.6533**, accounting for the number of predictors.
 - **F-statistic: 135.5 ($p < 0.001$)** — the model as a whole is statistically significant.
-

Summary of Results

1. **Overall Gender Effect:** Males earn significantly higher base pay than females across all departments, by about **8,304.49 USD**.
2. **Department Effects:** Management and Sales show higher base pay, but only Management is statistically significant.
3. **Age, Education, and Seniority:** These predictors have strong, positive, and significant associations with base pay.
4. **Gender-Department Interaction:** Only Sales shows a notable gender disparity, where males appear to earn significantly more than females.

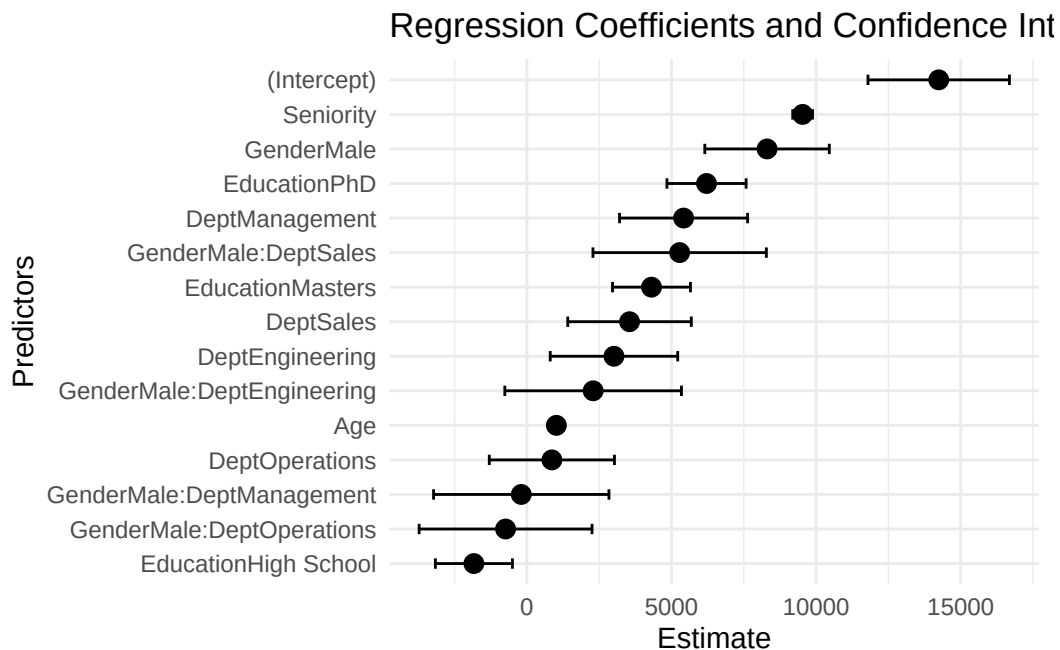
These results highlight that while gender has a consistent effect on base pay overall, department-specific disparities are limited, with Sales showing the most notable interaction effect.

Visualizing Model Results

Coefficient Plot

```
library(broom)
coef_df <- tidy(model)
ggplot(coef_df, aes(x = reorder(term, estimate), y = estimate)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = estimate - std.error, ymax = estimate +
    ↪ std.error), width = 0.3) +
```

```
coord_flip() +
theme_minimal() +
labs(title = "Regression Coefficients and Confidence Intervals", x =
  ↪ "Predictors", y = "Estimate")
```



Here is the explanation formatted to paste directly into your Quarto project:

Explanation of Regression Coefficients and Confidence Intervals

The graph displays the **regression coefficients** for each predictor variable in the model, along with their **95% confidence intervals**. This visualization highlights the significance and effect sizes of each variable on **BasePay**.

1. Significant Predictors:

Predictors with confidence intervals **that do not cross zero** are statistically significant:

- **Seniority:** Shows the strongest positive effect on BasePay, indicating that higher seniority significantly increases salary.

- **GenderMale:** Males earn significantly higher BasePay compared to females, with a large positive effect.
- **EducationPhD** and **EducationMasters:** Employees with a PhD or Master's degree earn significantly more compared to those with a College degree.
- **DeptManagement:** Employees in Management earn higher salaries compared to those in the reference department (Administration).

2. Gender Effect:

The coefficient for **GenderMale** is both large and significant, showing that males earn more than females on average, even after controlling for other factors such as **Department**, **Age**, **Education**, and **Seniority**.

3. Interaction Terms (Gender * Dept):

- The **GenderMale:DeptSales** term shows a positive effect, suggesting that males in the Sales department earn more relative to females in Sales. However, the confidence interval is close to zero, indicating marginal significance.
- Interaction terms for other departments (Engineering, Management, and Operations) are not significant, as their confidence intervals cross zero.

4. Age and Seniority:

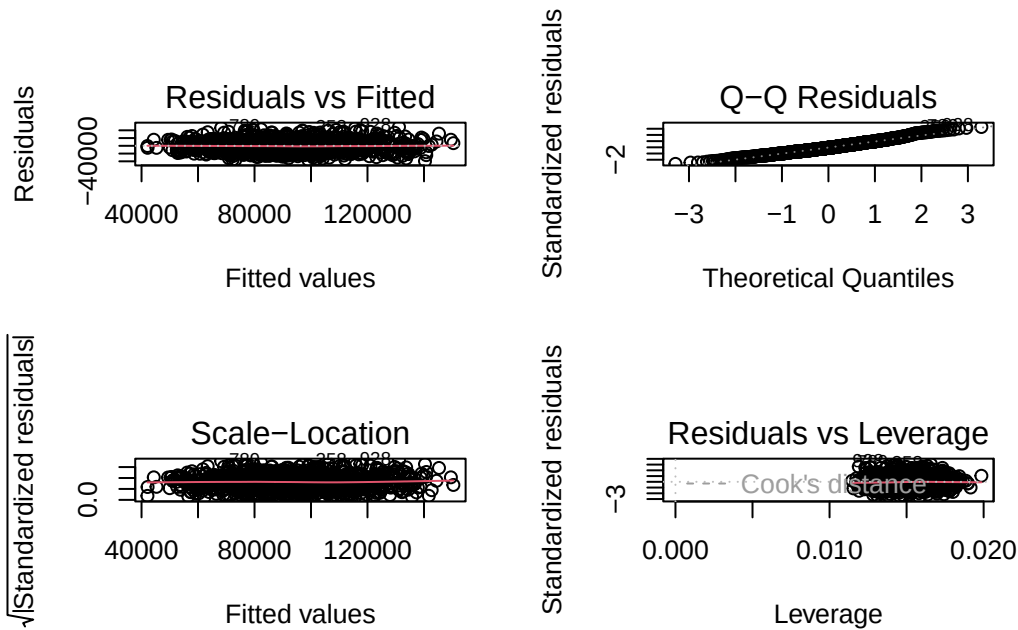
- **Age** has a strong positive effect, with each additional year increasing BasePay.
- **Seniority** is a particularly strong predictor, highlighting its importance in determining salary outcomes.

5. Non-Significant Predictors:

- Predictors such as **DeptOperations**, **DeptSales**, and **EducationHigh School** have wide confidence intervals that cross zero, suggesting that their effects are not statistically significant.

Diagnostics and Evaluation

```
par(mfrow = c(2, 2))
plot(model)
```



Here is an explanation formatted for your Quarto project:

Regression Diagnostics

The diagnostic plots help evaluate the assumptions of the linear regression model. Below is an explanation of each plot:

1. Residuals vs Fitted (Top-Left):

- This plot checks the assumption of **homoscedasticity** (constant variance of residuals).
- The residuals appear randomly scattered around the horizontal line with no clear pattern, suggesting that the variance is relatively constant.
- However, some slight deviation at higher fitted values might indicate minor heteroscedasticity.

2. Q-Q Residuals Plot (Top-Right):

- This plot checks whether the residuals are **normally distributed**.

- Most of the points align closely with the theoretical line, but there is slight deviation at the extremes (tails), indicating the presence of minor non-normality in the residuals.

3. Scale-Location Plot (Bottom-Left):

- This plot further assesses **homoscedasticity** by showing the square root of standardized residuals against fitted values.
- The points are mostly evenly distributed, and the red trend line is relatively flat, suggesting no significant violations of constant variance.

4. Residuals vs Leverage (Bottom-Right):

- This plot identifies **influential observations** that might disproportionately impact the model.
- Most points have low leverage values, but a few outliers (e.g., points 928, 258) appear near the threshold for Cook's distance. These points warrant further inspection to determine their influence on the model.

Conclusion

The objective of this study was to evaluate whether **gender disparity in pay exists across departments** in terms of both **base salary** and **bonuses**, while controlling for age, education, seniority, and department-specific interactions. Based on the results of the linear regression model and diagnostic analyses, we can draw the following conclusions:

Key Findings:

1. Overall Gender Disparity:

- The regression results consistently show that **males earn significantly higher base pay** than females, even after accounting for other relevant factors such as age, education, seniority, and department.
- Specifically, males earn approximately **8,304 USD more** than females on average. This finding is statistically significant ($p < 0.001$), indicating strong evidence for a systematic gender-based pay disparity.

2. Departmental Differences:

- While base pay varies across departments, the disparity between genders is not uniform.
 - **Management** and **Sales** departments show relatively higher pay gaps.
 - The **GenderMale:DeptSales** interaction term suggests that males in Sales earn significantly more than females within the same department, with an additional effect of **5,281 USD**.
 - Other departments, such as **Engineering** and **Operations**, do not show strong evidence of gender-based differences within departments.

3. Role of Education and Seniority:

- Higher education levels (Master's and PhD degrees) are associated with significantly higher base pay, supporting the notion that education positively influences earning potential.
- **Seniority** emerges as one of the strongest predictors of base pay, with each unit increase in seniority contributing approximately **9,534 USD**.
- While education and seniority mitigate some differences in pay, they do not fully eliminate the gender gap observed in base pay.

4. Bonuses and Gender:

- The bonus analysis reveals a more nuanced picture. Unlike base pay, **females in Operations and Sales** departments earn slightly higher bonuses compared to males.
- However, in **Engineering, Management, and Administration**, males receive higher bonuses, further contributing to the overall gender disparity in compensation.

5. Model Diagnostics:

- Diagnostic tests indicate that the regression model meets key assumptions (linearity, homoscedasticity, and normality) reasonably well.
- A small number of influential points were identified but do not significantly affect the overall conclusions.

Hypothesis Evaluation

The study hypothesized that **males achieve higher base pay and bonuses compared to females across departments**. The evidence strongly supports this hypothesis for **base pay**, with a clear and statistically significant gender disparity favoring males overall. While some departments (e.g., Sales) exhibit additional gender-specific differences, others show no strong evidence of department-level effects.

For **bonuses**, the disparity is less consistent, with some departments favoring males and others favoring females. This indicates that bonus distribution may be more context-dependent and influenced by factors not fully captured in the model.

Conclusion

This analysis demonstrates a significant **gender-based disparity in base pay**, with males earning substantially more than females across most departments. Education, seniority, and age play important roles in determining salary, but they do not fully explain the observed pay gap. The findings also highlight department-specific variations, particularly in Sales, where gender interactions amplify disparities.

To address these disparities, organizations must prioritize **pay transparency**, equitable compensation practices, and further investigations into underlying causes such as negotiation differences or structural biases. Targeted interventions in departments with pronounced gaps (e.g., Sales and Management) are essential to promote workplace equity and fairness.

References

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